

CSA15 Cloud Computing and Big Data Analytics

Capstone Cloud Technical Presentation

Individual Student Presentation Deck

Presentation Focus

Present your capstone as a cloud-backed product. Explain the problem, architecture, service choices, evidence, and CO1 learning clearly.

Presenter Details

Student Name: Shakthi M
Register Number:192511197
Class / Section / Slot: Slot C
Capstone Title: SleepScope:
Students Sleep and Stress
Analytics App
Selected SDG Goal(s):SDG 9
– Industry, Innovation and
Infrastructure
SDG 11 – Sustainable Cities and Communities
Cloud Platform / Tools:Google Firebase +
Google Cloud Platform (or AWS)

Problem Context and Stakeholder Mapping

Our capstone begins with a real problem to solve

Problem Statement

Students spend 15–30 minutes standing in queues during lunch breaks. Long waiting time, Food congestion, Poor crowd management, Order confusion.

Target Users

Students, Faculty, Canteen Staff, College Management

SDG Fit

Supports smart infrastructure and efficient campus services

Product Boundary

Online ordering, Live crowd indicator, Digital payment, Pickup token, Order tracking

Cloud Adoption Rationale and Concept Map

The problem needs cloud support to become a usable product

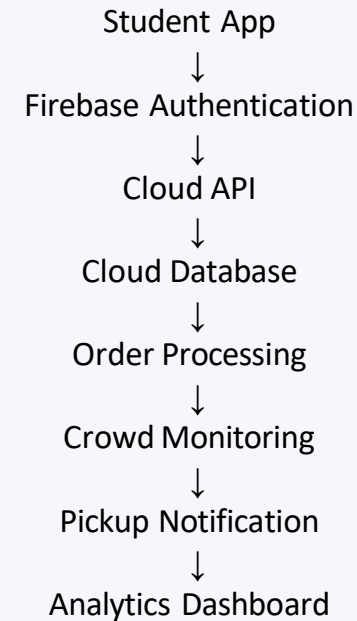
Cloud Definition

Cloud computing enables online ordering, data storage, authentication, and real-time updates accessible from anywhere..

Cloud Characteristics

Elasticity, scalability, measured service, resource pooling.

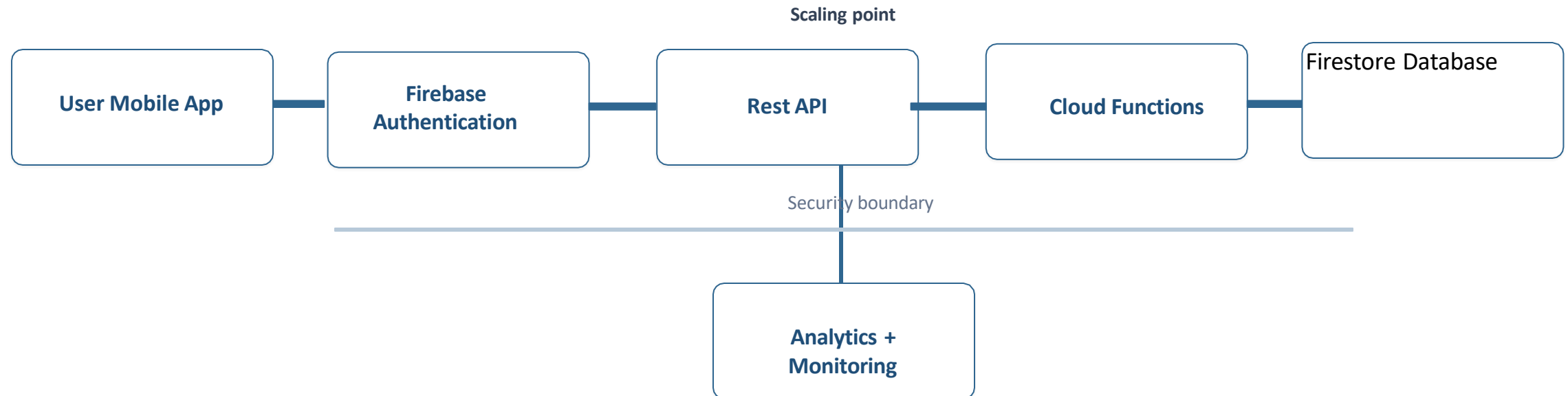
Cloud Concept Map



Guidance - delete before presentation: Use the concept map as a speaking aid. Do not overload it with tiny text.

Capstone Cloud Architecture

Our cloud architecture connects the user action to services



Guidance - delete before presentation: Explain the request path from user to cloud services. Replace sample labels with your actual architecture.

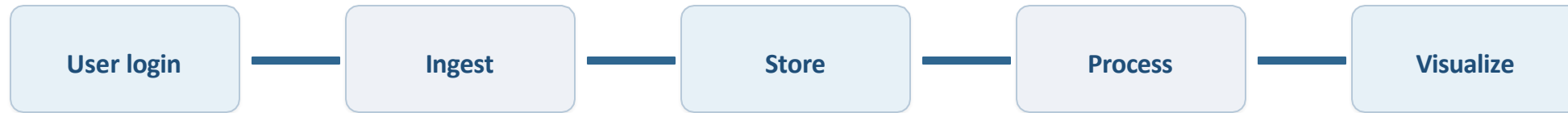
Cloud Service Model Responsibility Matrix

Each service model defines what we control and what cloud manages

Service area	Model	Provider manages	Student team manages		Why this fits	
Frontend hosting	Frontend Hosting	SaaS	Hosting	UI	Design	Easy deployment
Backend/API	Backend API	PaaS	Runtime	Busine	ss Logic	Fast development
Database	Database	DBaaS	Database	Dat	a Design	Real-time updates
Storage	Storage	IaaS	Storage	Upload Man	agement	Secure storage
Monitoring / logs	Monitoring	SaaS	Logs	Error	Analysis	Performance tracking
Security / IAM	Security	IAM	Authentication	Roles & Per	missions	Secure access

Data Flow and Analytics Pipeline

The product data moves from user activity to useful insight



Data Requirements

User data, events, logs, files, sensor data, or transactions.

Analytics Requirement

Dashboard, alert, prediction, report, trend, or decision support.

Proof Required

Sample input, sample output, and one insight generated.

Python / AI Module Integration

Python or AI adds intelligence where the product needs it

Python Role

Crowd prediction,Waiting time estimation,Sales analytics.

Input and Output

Previous orders,Current queue,Time

Output

Estimated waiting time

Crowd level

Execution Point

Cloud Function

Product Value

Students know the best pickup time before visiting the canteen

Security, Scalability, Privacy and Cost Design

Security, scale, privacy and cost shape the final design

Security

Firebase Authentication, HTTPS, Role-based login

Privacy

Encrypted user information, Secure payment data, Limited access.

Scalability

Supports thousands of concurrent users using cloud auto-scaling.

Cost Awareness

Uses Firebase Free Tier with low operational cost.

Implementation Stack and Deployment Evidence

Our implementation stack shows how the design will be built

Mobile / Web Layer

Android, iOS, Flutter, React, or web frontend.

Cloud Services

Compute, database, storage, authentication, analytics, API tools.

Evidence

GitHub, API test, cloud dashboard, endpoint, logs, or deployment URL.

Version Control

Repository structure, commits, branches, and README clarity.

Demo Storyboard and User Journey

A complete user journey proves the product flow

Scene 1

Student logs in
↓
Authentication successful
↓
Firebase Authentication

Scene 2

Student selects food
↓
Menu loads
↓
Firestore

Scene 3

Student places order
↓
API processes order
↓
Cloud Function

Scene 4

Payment successful
↓
Database updated
↓
Firestore

Scene 5

Crowd level displayed
↓
Analytics service

Scene 6

Pickup notification sent
↓
Firebase Cloud Messaging

Evaluation Readiness Checklist

Our evidence confirms that the presentation is ready

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Cloud ecosystem and service model explained

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Elasticity and scalability decisions are visible

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REST/SOAP/API role connected to capstone

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GitHub and cloud evidence are inspectable

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Architecture has compute, data, storage, security, monitoring

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References and integrity statement are included

Evaluator Focus

CO1 is verified through cloud design reasoning, service choices, evidence, and individual explanation.

Conclusion, Limitations and Future Scope

We close with learning, limitations and next steps

Learning Outcome

Understood cloud computing Implemented cloud architecture
Used APIs Learned scalable application developmen

Limitations

Currently designed for one canteen Basic AI prediction Limited
payment options

Future Improvement

AI-based demand forecasting Multi-campus deployment Face
recognition pickup IoT-based smart counters

Integrity Statement

Project developed using Firebase, Flutter, GitHub, and AI tools
for documentation assistance.